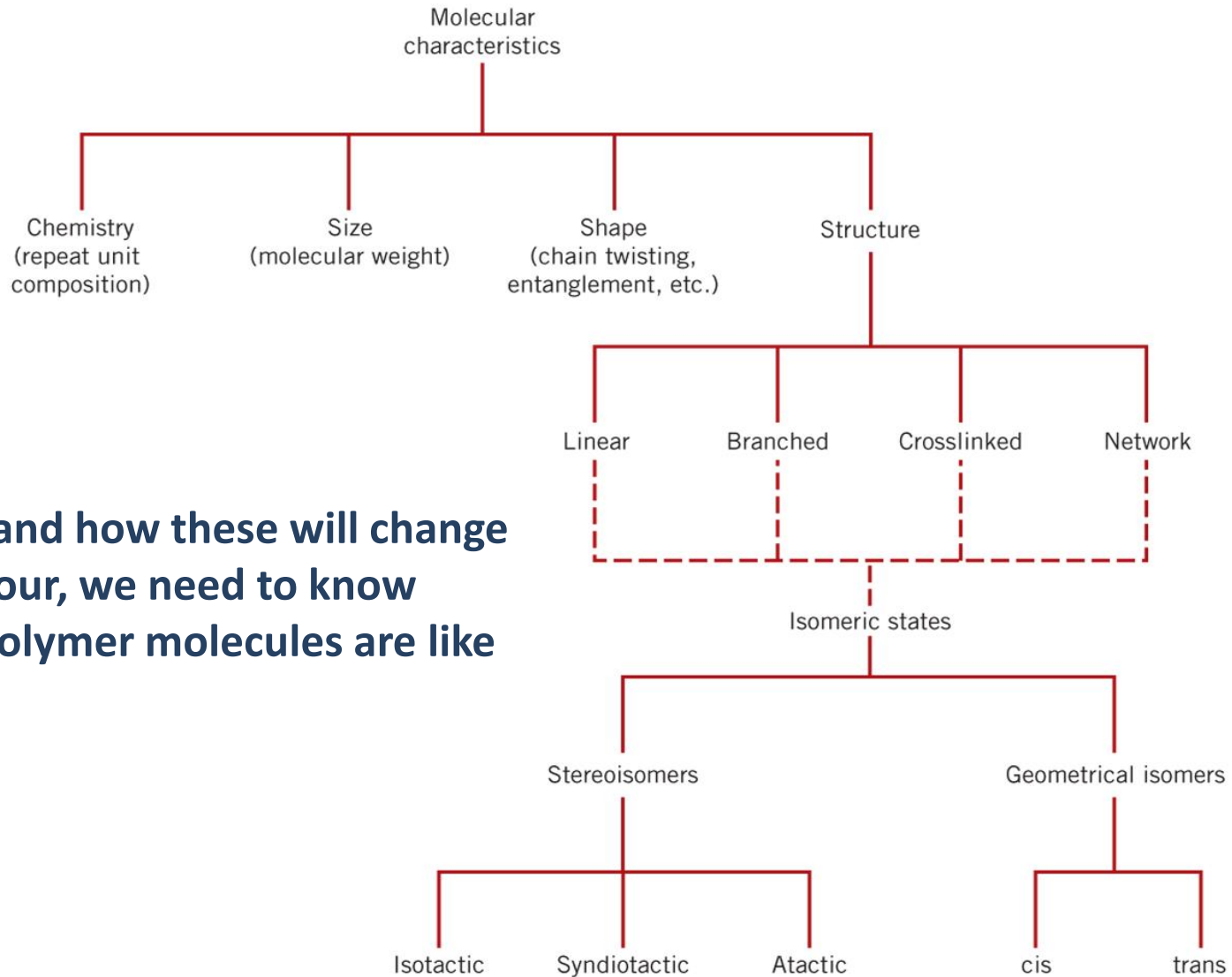




Polymers and Elastomers

Presented by Dr Adijat 'Wumi Inyang

Polymer characteristics



To understand how these will change the behaviour, we need to know what the polymer molecules are like

Fig 5.8 Callister & Rethwisch 9e

Common polymers

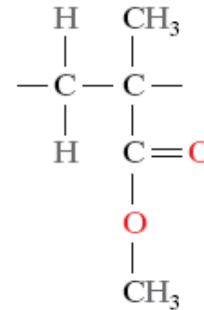
<i>Polymer</i>	<i>Repeat Unit</i>	
Polyethylene (PE)	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{H} \end{array}$	No side-group but can branch Low T_g
Poly(vinyl chloride) (PVC)	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{Cl} \end{array}$	Polar side group High T_g
Polytetrafluoroethylene (PTFE)	$\begin{array}{c} \text{F} \quad \text{F} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{F} \quad \text{F} \end{array}$	Highly symmetrical High T_m
Polypropylene (PP)	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{CH}_3 \end{array}$	Non-polar side group
Polystyrene (PS)	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{C}_6\text{H}_5 \end{array}$	Bulky side group High T_g

Common polymers

Polymer

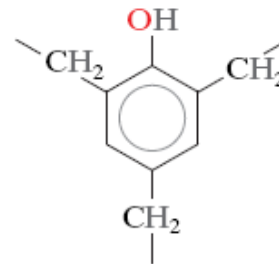
Repeat Unit

Poly(methyl methacrylate) (PMMA)



Large and polar side group
 $T_g = 105^\circ\text{C}$

Phenol-formaldehyde (Bakelite)



Thermoset

Common polymers

Polymer	Repeat Unit	
Poly(hexamethylene adipamide) (nylon 6,6)	$\text{---N---}\left[\begin{array}{c} \text{H} \\ \\ \text{---C---} \\ \\ \text{H} \end{array}\right]_6\text{---N---}\overset{\text{O}}{\parallel}\text{C---}\left[\begin{array}{c} \text{H} \\ \\ \text{---C---} \\ \\ \text{H} \end{array}\right]_4\text{---}\overset{\text{O}}{\parallel}\text{C---}$	Large T range between T_g and T_m
Poly(ethylene terephthalate) (PET, a polyester)	$\text{---}\overset{\text{O}}{\parallel}\text{C---}\overset{a}{\text{C}_6\text{H}_4}\text{---}\overset{\text{O}}{\parallel}\text{C---O---}\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{---C---C---} \\ \quad \\ \text{H} \quad \text{H} \end{array}\text{---O---}$	
Polycarbonate (PC)	$\text{---O---}\overset{a}{\text{C}_6\text{H}_4}\text{---}\begin{array}{c} \text{CH}_3 \\ \\ \text{C} \\ \\ \text{CH}_3 \end{array}\text{---}\text{C}_6\text{H}_4\text{---O---}\overset{\text{O}}{\parallel}\text{C---}$	Stiff backbone High T_g

Common elastomers

<i>Repeat Unit Name</i>	<i>Repeat Unit Structure</i>	<i>Repeat Unit Name</i>	<i>Repeat Unit Structure</i>
Acrylonitrile	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{C}\equiv\text{N} \end{array} $	Isoprene	$ \begin{array}{c} \text{H} \quad \text{CH}_3 \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ -\text{C}-\text{C}=\text{C}-\text{C}- \\ \quad \quad \quad \\ \text{H} \quad \quad \quad \text{H} \end{array} $
Styrene	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{C}_6\text{H}_5 \end{array} $	Isobutylene	$ \begin{array}{c} \text{H} \quad \text{CH}_3 \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{CH}_3 \end{array} $
Butadiene	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ -\text{C}-\text{C}=\text{C}-\text{C}- \\ \quad \quad \quad \\ \text{H} \quad \quad \quad \text{H} \end{array} $	Dimethylsiloxane	$ \begin{array}{c} \text{CH}_3 \\ \\ -\text{Si}-\text{O}- \\ \\ \text{CH}_3 \end{array} $
Chloroprene	$ \begin{array}{c} \text{H} \quad \text{Cl} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ -\text{C}-\text{C}=\text{C}-\text{C}- \\ \quad \quad \quad \\ \text{H} \quad \quad \quad \text{H} \end{array} $		

Applications of polymers

<i>Material Type</i>	<i>Trade Names</i>	<i>Major Application Characteristics</i>	<i>Typical Applications</i>
<i>Thermoplastics</i>			
Acrylonitrile-butadiene-styrene (ABS)	Abson Cyclocac Kralastic Lustran Lucon Novodur	Outstanding strength and toughness, resistant to heat distortion; good electrical properties; flammable and soluble in some organic solvents	Under-the-hood automotive applications, refrigerator linings, computer and television housings, toys, highway safety devices
Acrylics [poly(methyl methacrylate)]	Acrylite Diakon Lucite Paraloid Plexiglas	Outstanding light transmission and resistance to weathering; only fair mechanical properties	Lenses, transparent aircraft enclosures, drafting equipment, bathtub and shower enclosures
Fluorocarbons (PTFE or TFE)	Teflon Fluon Halar Hostaflon TF Neoflon	Chemically inert in almost all environments, excellent electrical properties; low coefficient of friction; may be used to 260°C; relatively weak and poor cold-flow properties	Anticorrosive seals, chemical pipes and valves, bearings, wire and cable insulation, antiadhesive coatings, high-temperature electronic parts
Polyamides (nylons)	Nylon Akulon Durethan Fostamid Nomex Ultramid Zytel	Good mechanical strength, abrasion resistance, and toughness; low coefficient of friction; absorbs water and some other liquids	Bearings, gears, cams, bushings, handles, and jacketing for wires and cables, fibers for carpet, hose, and belt reinforcement

High toughness

Glassy

High T_m
Low friction

High toughness and wear resistance

Applications of polymers

<i>Material Type</i>	<i>Trade Names</i>	<i>Major Application Characteristics</i>	<i>Typical Applications</i>
Thermoplastics			
Polycarbonates	Calibre Iupilon Lexan Makrolon Novarex	Dimensionally stable; low water absorption; transparent; very good impact resistance and ductility; chemical resistance not outstanding	Safety helmets, lenses, light globes, base for photographic film, automobile battery cases
Polyethylenes	Alathon Alkathene Fortiflex Hifax Petrothene Rigidex Zemid	Chemically resistant and electrically insulating; tough and relatively low coefficient of friction; low strength and poor resistance to weathering	Flexible bottles, toys, tumblers, battery parts, ice trays, film wrapping materials, automotive gas tanks
Polypropylenes	Hicor Meraklon Metocene Polypro Pro-fax Propak Propathene	Resistant to heat distortion; excellent electrical properties and fatigue strength; chemically inert; relatively inexpensive; poor resistance to ultraviolet light	Sterilizable bottles, packaging film, automotive kick panels, fibers, luggage
Polystyrenes	Avantra Dylene Innova Lutex Styron Vestylon	Excellent electrical properties and optical clarity; good thermal and dimensional stability; relatively inexpensive	Wall tile, battery cases, toys, indoor lighting panels, appliance housings, packaging

High impact resistance

Low T_g
Chemically inert

Easily moulded
Fatigue resistant
Chemically inert

Glassy
Insulating

Applications of polymers

<i>Material Type</i>	<i>Trade Names</i>	<i>Major Application Characteristics</i>	<i>Typical Applications</i>
<i>Thermoplastics</i>			
Vinyls	Dural Formolon Geon Pevikon Saran Tygon Vinidur	Good low-cost, general-purpose materials; ordinarily rigid, but may be made flexible with plasticizers; often copolymerized; susceptible to heat distortion	Floor coverings, pipe, electrical wire insulation, garden hose, shrink wrap
Polyesters (PET or PETE)	Crystar Dacron Eastapak HiPET Melinex Mylar Petra	One of the toughest of plastic films; excellent fatigue and tear strength, and resistance to humidity, acids, greases, oils, and solvents	Oriented films, clothing, automotive tire cords, beverage containers

Easily moulded
Cheap

High toughness
Chemically resistant

Applications of polymers

<i>Material Type</i>	<i>Trade Names</i>	<i>Major Application Characteristics</i>	<i>Typical Applications</i>
<i>Thermosetting Polymers</i>			
Epoxies	Araldite Epikote Lytex Maxive Sumilite Vipel	Excellent combination of mechanical properties and corrosion resistance; dimensionally stable; good adhesion; relatively inexpensive; good electrical properties	Electrical moldings, sinks, adhesives, protective coatings, used with fiberglass laminates
Phenolics	Bakelite Duralite Milex Novolac Resole	Excellent thermal stability to more than 150°C; may be compounded with a large number of resins, fillers, etc.; inexpensive	Motor housings, adhesives, circuit boards, electrical fixtures
Polyesters	Aropol Baygal Derakane Luytex Vitel	Excellent electrical properties and low cost; can be formulated for room- or high-temperature use; often fiber reinforced	Helmets, fiberglass boats, auto body components, chairs, fans

Source: Adapted from C. A. Harper (Editor), *Handbook of Plastics and Elastomers*. Copyright © 1975 by McGraw-Hill Book Company. Reproduced with permission.

All have excellent stability and adhesive properties
All can be used as matrix in composite (hybrid) materials

Applications of elastomers

<i>Chemical Type</i>	<i>Trade (Common) Names</i>	<i>Elongation (%)</i>	<i>Useful Temperature Range (°C)</i>	<i>Major Application Characteristics</i>	<i>Typical Applications</i>
Natural polyisoprene	Natural rubber (NR)	500–760	–60 to 120	Excellent physical properties; good resistance to cutting, gouging, and abrasion; low heat, ozone, and oil resistance; good electrical properties	Pneumatic tires and tubes; heels and soles; gaskets; extruded hose
Styrene-butadiene copolymer	GRS, Buna S (SBR)	450–500	–60 to 120	Good physical properties; excellent abrasion resistance; not oil, ozone, or weather resistant; electrical properties good, but not outstanding	Same as natural rubber
Acrylonitrile-butadiene copolymer	Buna A, Nitrile (NBR)	400–600	–50 to 150	Excellent resistance to vegetable, animal, and petroleum oils; poor low-temperature properties; electrical properties not outstanding	Gasoline, chemical, and oil hose; seals and O-rings; heels and soles; toys
Chloroprene	Neoprene (CR)	100–800	–50 to 105	Excellent ozone, heat, and weathering resistance; good oil resistance; excellent flame resistance; not as good in electrical applications as natural rubber	Wire and cable; chemical tank linings; belts, hoses, seals, and gaskets
Polysiloxane	Silicone (VMQ)	100–800	–115 to 315	Excellent resistance to high and low temperatures; low strength; excellent electrical properties	High- and low-temperature insulation; seals, diaphragms; tubing for food and medical uses

Natural rubber

Synthetic rubber

Co-polymer elastomer

Good chemical resistance

Thermal stability

Advanced polymers

Ultrahigh Molecular Weight Polyethylene (UHMWPE) – Spectra™



1. Extremely high impact resistance
2. Outstanding resistance to wear and abrasion
3. A very low coefficient of friction
4. A self-lubricating and non-stick surface
5. Very good chemical resistance to normally encountered solvents
6. Excellent low-temperature properties
7. Outstanding sound damping and energy absorption characteristics
8. Electrically insulating and excellent dielectric properties

Advanced polymers

Liquid Crystal polymers



Rod-shaped rigid molecules

Not described by conventional ideas of liquid, amorphous, crystalline or semi-crystalline state



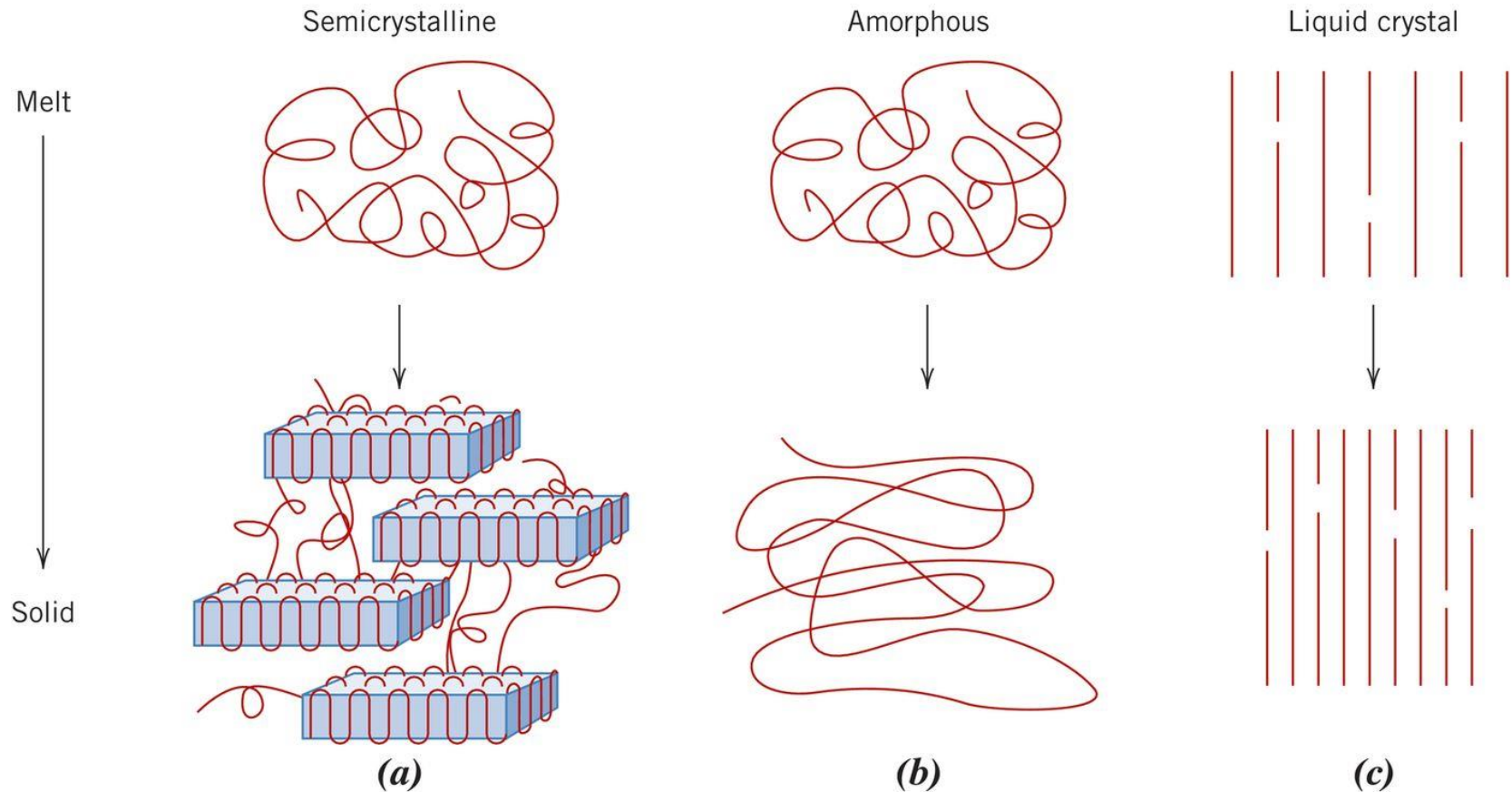
Different state – **liquid crystals**

Good thermal stability and chemically inert

Arranged in electric field to make displays

Advanced polymers

Molecular structures from liquid to solid state



Adapted from G. W. Calundann and M. Jaffe, "Anisotropic Polymers, Their Synthesis and Properties," Chapter VII in Proceedings of the Robert A. Welch Foundation Conferences on Polymer Research, 26th Conference, Synthetic Polymers, Nov. 1982.

Advanced polymers

Thermoplastic elastomers

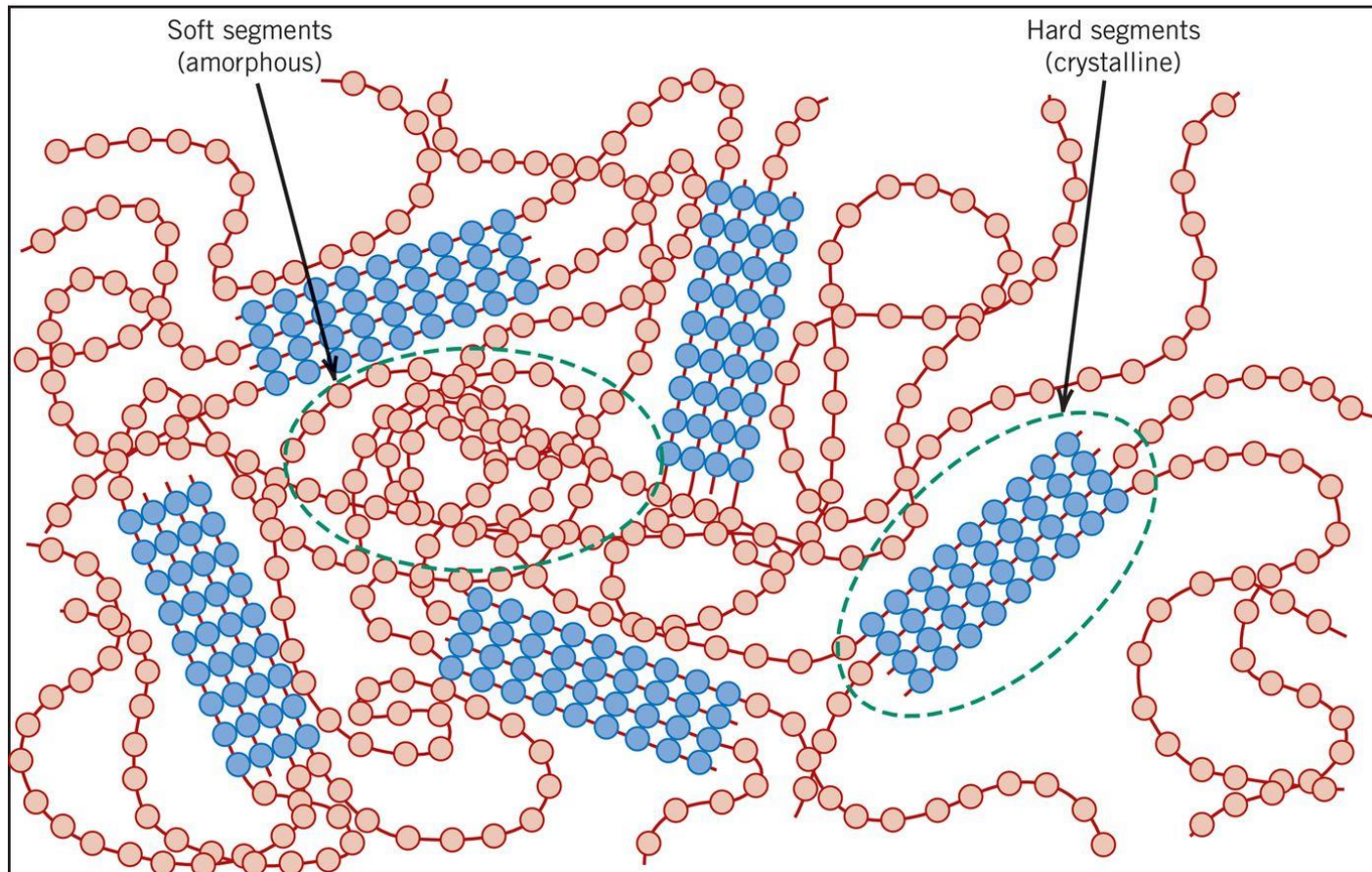


1. Rubbery behavior at room temperature yet is thermoplastic
2. Can be melted from hard phase to be processed by conventional forming techniques
3. Melting-solidification is reversible
4. It can be re-shaped

Advanced polymers

Thermoplastic elastomers

block co-polymers e.g. polyurethanes



Summary

From the chemistry and structure of the polymers we can understand their:

- Thermal behaviour
- Applications
- How they will change with processing